



Carlo F. Bernardez

Mechanical Engineering Student

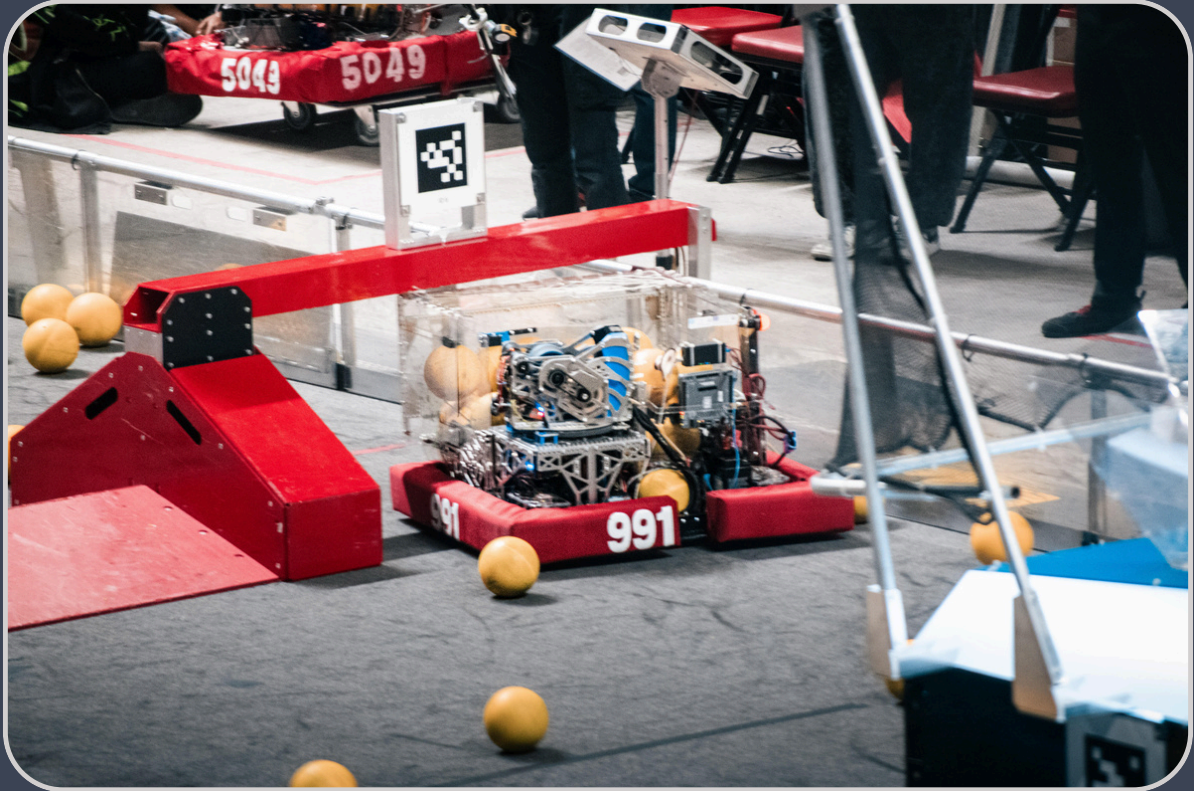
Arizona State University
Barrett, The Honors College

Engineering Portfolio | 2026



I am a Mechanical Engineering student at Arizona State University with interests in robotics, mechanical design, manufacturing, and product development. I enjoy transforming ideas into functional systems through CAD, fabrication, electrical integration, and hands-on prototyping. Through robotics and engineering projects, I have developed experience in multidisciplinary collaboration, manufacturing, and technical problem-solving. As I continue my engineering education, I look forward to expanding my skills through challenging coursework, personal projects, and industry experience while contributing to innovative engineering teams.

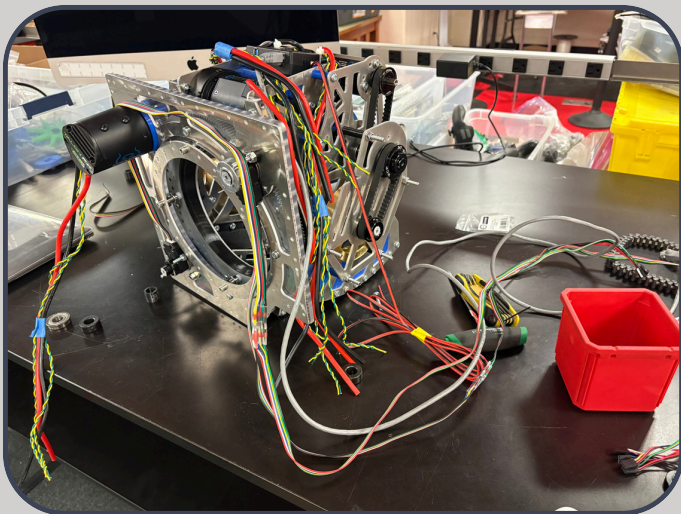
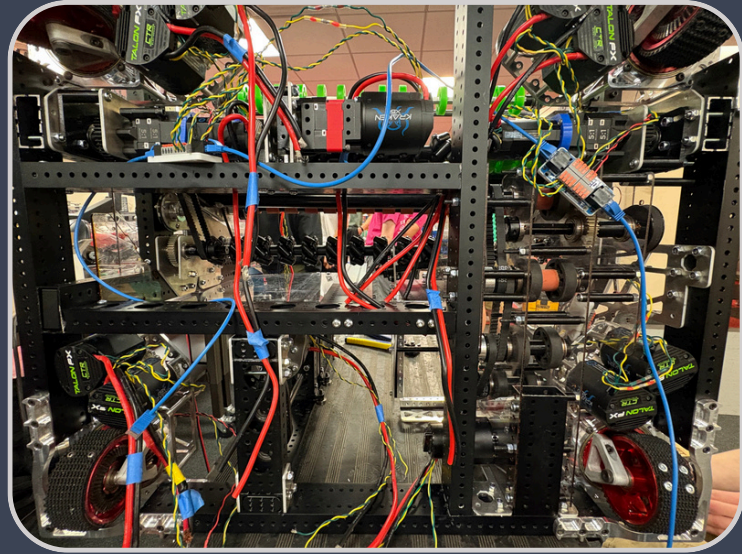
FRC Competition Robot



The FIRST Robotics Competition (FRC) is an international engineering competition that challenges teams to design, manufacture, and program a fully functional robot within a limited build season. As Team Operations Director and President of Brophy Robotics, I contributed to the development of our competition robot through electrical system integration, mechanical assembly, fabrication, and iterative testing. This project provided hands-on experience with multidisciplinary engineering while emphasizing reliability, manufacturability, and effective teamwork.

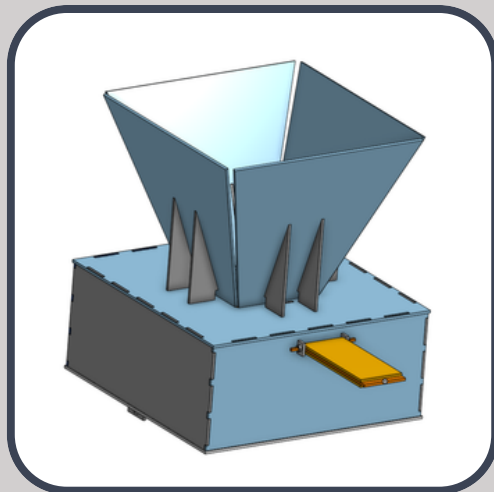
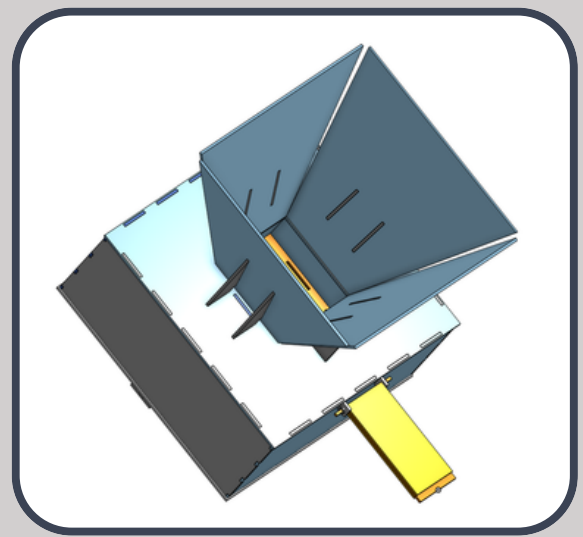
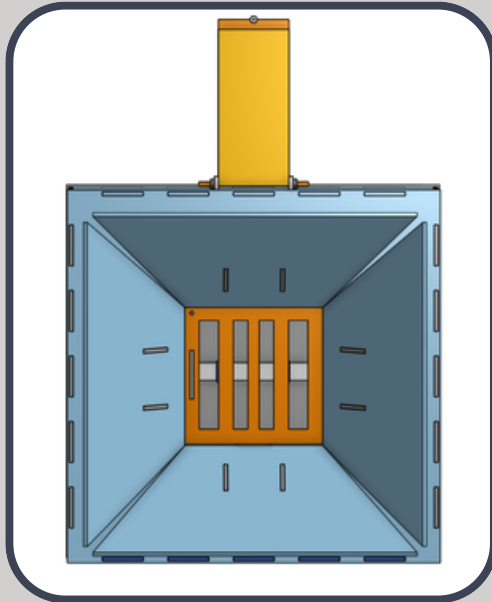
Electrical Integration

Power and CAN bus routing for the robot drivetrain during the integration phase. Wiring was organized and secured to provide reliable communication between motor controllers and subsystems while minimizing interference, simplifying maintenance, and improving accessibility throughout the competition season.



Electrical integration of the robot's rotating turret assembly. Power and CAN bus wiring were routed to support continuous turret movement while maintaining reliable communication, minimizing cable strain, and allowing for efficient maintenance and troubleshooting.

Solar Oven



Objective

Design a compact, low-cost solar oven that maximized heat retention and solar energy capture while remaining manufacturable within project constraints. The final design incorporated reflective surfaces, insulated walls, adjustable sun tracking, and an Arduino-based temperature monitoring system.

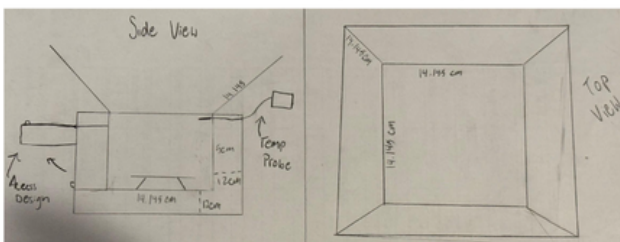
My Contributions

- Assisted with CAD development in Onshape
- Integrated the Arduino-based temperature monitoring system
- Supported laser cutting and fabrication
- Assisted in prototype and final assembly
- Conducted outdoor testing and performance analysis

Design Highlights

- Dual-layer Mylar glazing
- Adjustable stand for solar tracking
- Newspaper insulation
- Arduino temperature monitoring

Design Parameters	Units	Combo1	C
L (Length of Oven Window)	cm	14.145	
W (Width of Oven Window)	cm	14.145	
h (Height of Oven Window)	cm	5	
n (Number of layers of Mylar)		2	
Volume	cm ³	1000	



Material	Individual	Cost
Mylar	0.25 each	0.5
Inner Box	1.75/m ²	0.1197
Outer Box	1.75/m ²	1.105857
Labor	2 per hr	10
transportation	0.20 per lb	
reflectors	1.75/m ²	0.4508
Total Cost		12.176357

Cardboard Canoe



Objective

Design and construct a lightweight, structurally sound cardboard canoe capable of remaining buoyant while meeting competition constraints. The design process incorporated mathematical hull modeling, manufacturing planning, and dimensional verification before fabrication.

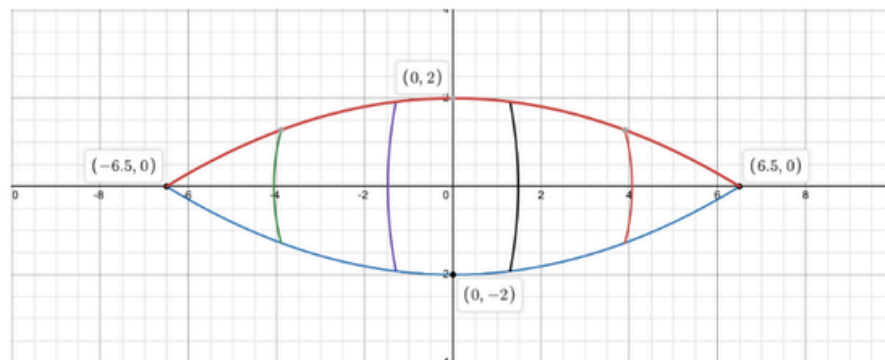
My Contributions

- Assisted with hull design and dimensional planning
- Supported fabrication and structural assembly
- Manufactured cardboard components using laser cutting and manual fabrication
- Verified dimensions and construction tolerances
- Participated in final assembly and competition testing

Design Highlights

- Mathematical hull geometry
- Laser-cut structural components
- Lightweight reinforced construction
- Dimensional verification after fabrication

Top View:



(Red Line) $y = -0.0473x^2 + 2 \{ |x| \leq 6.5 \}$

(Blue Line) $y = 0.0473x^2 - 2 \{ |x| \leq 6.5 \}$

6.62403

Thwart size:

- Carpet tubes 4.5 Diameter
- Or Three inches equilateral triangle
- 4 Thwarts at $x = \pm 3.9$ feet and ± 1.3 feet
 - Thwart 1 (Stern-most): $x = 0.1y^2 - 4.066 \{ |y| \leq 1.28 \}$
 - Thwart 2 (Aft-center): $x = 0.05y^2 - 1.48 \{ |y| \leq 1.92 \}$
 - Thwart 3 (Bow-center): $x = -0.05y^2 + 1.48 \{ |y| \leq 1.92 \}$
 - Thwart 4 (Bow-most): $x = -0.1y^2 + 4.066 \{ |y| \leq 1.28 \}$